

One-Dimensional Ising Model

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The Model and Hamiltonian

We consider a one-dimensional chain of N sites. For periodic boundary conditions, we impose that site $N + 1$ is equivalent to site 1 ($N + i \equiv i$).

The Hamiltonian for the system is given by:

$$H = -J \sum_{i=1}^N s_i s_{i+1} - h \sum_{i=1}^N s_i \quad (1)$$

where $s_i = \pm 1$, J is the coupling constant, and h is the external magnetic field.

An alternative form of the Hamiltonian, useful for the transfer matrix method, is:

$$H = -J \sum_{i=1}^N s_i s_{i+1} - \frac{h}{2} \sum_{i=1}^N (s_i + s_{i+1}) \quad (2)$$

Partition Function and Transfer Matrix

The partition function Z is the trace of the Boltzmann factor, with $\beta = 1/(k_B T)$:

$$\begin{aligned} Z &= \text{Tr } e^{-\beta H} \\ &= \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \exp \left[\beta \sum_{i=1}^N \left(J s_i s_{i+1} + \frac{h}{2} (s_i + s_{i+1}) \right) \right] \\ &= \sum_{s_1} \cdots \sum_{s_N} \prod_{i=1}^N \exp \left[\beta \left(J s_i s_{i+1} + \frac{h}{2} (s_i + s_{i+1}) \right) \right] \end{aligned}$$

We define a **transfer matrix** T with elements $\langle s_i | T | s_{i+1} \rangle$:

$$\langle s | T | s' \rangle = \exp \left[\beta \left(J s s' + \frac{h}{2} (s + s') \right) \right] \quad (3)$$

Using this, the partition function becomes:

$$Z = \sum_{s_1} \cdots \sum_{s_N} \langle s_1 | T | s_2 \rangle \langle s_2 | T | s_3 \rangle \cdots \langle s_N | T | s_1 \rangle = \sum_{s_1} \langle s_1 | T^N | s_1 \rangle = \text{Tr}(T^N)$$

Matrix Elements and Eigenvalues

The elements of the 2×2 transfer matrix T are:

$$\begin{aligned} \langle + | T | + \rangle &= e^{\beta(J+h)} \\ \langle - | T | - \rangle &= e^{\beta(J-h)} \\ \langle + | T | - \rangle &= \langle - | T | + \rangle = e^{-\beta J} \end{aligned}$$

So, the transfer matrix is:

$$T = \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix} \quad (4)$$

To find the eigenvalues λ , we solve the characteristic equation $\det(T - \lambda I) = 0$:

$$\left(e^{\beta(J+h)} - \lambda \right) \left(e^{\beta(J-h)} - \lambda \right) - e^{-2\beta J} = 0$$

This expands to the quadratic equation:

$$\lambda^2 - \lambda \left(e^{\beta(J+h)} + e^{\beta(J-h)} \right) + e^{2\beta J} - e^{-2\beta J} = 0$$

$$\lambda^2 - 2\lambda e^{\beta J} \cosh(\beta h) + 2 \sinh(2\beta J) = 0$$

Solving for λ using the quadratic formula gives the two eigenvalues, λ_+ and λ_- :

$$\lambda_{\pm} = e^{\beta J} \left[\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right] \quad (5)$$

Thermodynamic Properties

Free Energy

The partition function is $Z = \lambda_+^N + \lambda_-^N$. In the thermodynamic limit ($N \rightarrow \infty$), the larger eigenvalue λ_+ dominates, so $Z \approx \lambda_+^N$.

The Helmholtz free energy per site is:

$$\begin{aligned} f &= -\frac{1}{N} k_B T \ln Z \approx -k_B T \ln \lambda_+ \\ &= -k_B T \ln \left(e^{\beta J} \left[\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right] \right) \\ &= -J - k_B T \ln \left[\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right] \end{aligned}$$

Magnetization

The magnetization per site, m , is found by differentiating the free energy with respect to the magnetic field h :

$$\begin{aligned} m &= -\frac{\partial f}{\partial h} = k_B T \frac{\partial}{\partial h} \ln \left[\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right] \\ &= \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}} \end{aligned}$$

Behavior of Magnetization

